

Emergency Response Drones

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Abstract—This article investigates the deployment of autonomous unmanned aerial vehicles (UAVs) operating as cooperative swarms to enhance real-time disaster response capabilities. By integrating a decentralized swarm intelligence algorithm—adapted from bio-inspired models such as Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO)—the proposed system enables drones to self-organize, distribute tasks dynamically, and maintain robust communication in uncertain and GPS-denied environments. The study presents a simulation evaluation of swarm-based fire detection, mapping, and suppression workflows, comparing them against conventional single-drone and centrally controlled multi-UAV methods. Results indicate that swarming algorithms significantly reduce response latency, improve area coverage efficiency by up to 40%, and increase system resilience against individual drone failure. The findings demonstrate the feasibility of swarm-enabled UAV systems as a scalable and cost-effective asset for first responders, with implications for future integration into smart-city emergency management infrastructures.

Index Terms—Drone swarm, fire detection, autonomous systems, thermal sensing, cooperative robotics, emergency response, PSO, ACO

I. INTRODUCTION

During disaster response, quick and precise situational assessment is essential, but conventional search-and-rescue techniques are constrained by slow deployment, restricted access to dangerous locations, and significant risk to responders. Although the usage of unmanned aerial vehicles (UAVs) to support emergency operations is growing, most current systems rely on either centralized multi-UAV control or single-drone missions, both of which have limitations regarding scalability, communication, and reliability. A possible solution is offered by swarm robotics, which uses decentralized algorithms to allow several UAVs to cooperate independently. Swarm systems, which draw inspiration from collective biological behaviours, can distribute duties, adjust to changing circumstances, and carry on even when individual drones fail. Nevertheless, there is still little confirmation outside of simulation for the use of swarm algorithms in real-time emergency response. An autonomous UAV swarm structure intended for emergency situations is proposed and assessed in this paper. The system allows for cooperative search, mapping, and victim location without relying on a central controller through the use of a decentralized swarming algorithm. When compared to traditional UAV deployment methods, experimental results show advantages in coverage efficiency, response time, and system robustness.

II. LITERATURE REVIEW AND STATE-OF-THE-ART

A. Software Platform

The swarm system was developed and tested within the **Unity Engine**, chosen for its robust physics modelling, lighting, and path-finding features. C# scripts implement the Ant Colony Optimization (ACO) swarm logic, communication protocols, and sensory emulation. Each UAV is instantiated as a separate agent object executing its own decision cycle. Unity's modular environment allows dynamic spawning of obstacles, environmental hazards, and victims, enabling repeatable disaster simulations.

B. Communication framework

Peer-to-peer communication among UAVs is simulated using a mesh network topology. Each drone maintains a local broadcast radius of 200 m with a simulated 5 % packet loss to emulate real-world interference. When within range, UAVs exchange pheromone-map data and victim alerts using lightweight UDP-style message passing. A base station serves as a passive monitor, receiving periodic mission summaries and detected victim coordinates for visualization, but does not directly control UAVs.

C. Data Management and Integration

All sensor readings, position data, and pheromone updates are logged to a shared database within Unity for analysis. The simulation results are exported to CSV files for post-processing using Python and MATLAB, allowing quantitative evaluation of coverage, latency, and robustness metrics.

III. SYSTEM OVERVIEW AND ARCHITECTURE

A. System Overview

The proposed system implements an autonomous, two-tier swarm network of unmanned aerial vehicles (UAVs) for early wildfire detection and suppression. The swarm comprises two types of agents: scout drones and payload drones. Each drone operates under a decentralized coordination model inspired by PSO and ACO principles, allowing the swarm to perform large-scale surveillance and cooperative firefighting without reliance on a centralized controller [1]. The swarm's primary goals are to (1) maximize terrain coverage, (2) minimize time-to-victim detection, and (3) maintain resilience to communication loss and drone failure.

The architecture incorporates three key bio-inspired design principles:

1. **Self-Organization (PSO-inspired)**: Similar to how particles in PSO optimize placements based on individual and global optimal solutions, drones autonomously modify their positions and behaviours

depending on swarm-level information and local observations.

2. **Stigmergic Communication (ACO-inspired):** Fire detection events function as environmental markers (analogous to pheromone trails) that trigger coordinated swarm responses without requiring continuous centralized commands.
3. **Emergent Coordination:** Simple local rules carried out by individual drones give rise to complex suppression behaviours that allow the swarm to adjust to changing fire circumstances without the need for explicit global planning.

Scout drones are equipped with thermal sensors and visual cameras to continuously patrol a designated area. When a scout drone detects an abnormal rise in temperature, it transitions from patrol mode to investigation mode, navigating toward the hotspot to verify the presence of an active fire.

B. Detection and Verification Process

The detection phase relies on thermal data thresholds and pattern recognition to filter out false positives such as sunlight reflections or heated ground surfaces. The **thermal beacon radius** R_{beacon} scales dynamically with fire **intensity**, (I).

- $R_{\text{beacon}} = R_{\text{base}} + k \cdot I$

where R_{base} is the **minimum detection range** and k is an **intensity scaling factor**.

Once a hotspot is identified, the drone performs a localized investigation, hovering above the source to gather additional temperature and image data. The drone collects thermal readings and applies a confidence estimation algorithm:

- $C_{\text{fire}} = I \cdot (0.6 + 0.4 \cdot p)$

where C_{fire} is the **fire confidence score**, (I) is the **normalized intensity**, and p is the **proximity factor** computed as:

- $p = 1 - (d / R_{\text{beacon}})$

If the onboard classifier confirms a fire source = true (when C_{fire} exceeds verification threshold), the drone transmits the geolocation, temperature readings, and visual confirmation to the swarm control network. This triggers the activation of the payload drone cluster for firefighting deployment.

C. Payload Dribe Swarm Response

The payload drones execute a coordinated suppression response inspired by PSO principles. Upon receiving the fire coordinates, the drones form an optimized formation to maximize coverage and minimize overlap. Each drone is equipped with foam sprayers or water-based extinguishing payloads.

The formation control algorithm adapts PSO principles where each drone optimizes its position based on:

1. **Personal objective:** Maintain effective suppression range from fire
2. **Swarm objective:** Distribute evenly around fire perimeter to maximize coverage
3. **Collision avoidance:** Maintain safe separation from other drones and obstacles

Deployed payload drones establish an orbital formation around the fire source using distributed slot assignment:

- $\theta_i = \theta_{\text{base}} + (360^\circ / N_{\text{deploy}}) \cdot i$

where θ_i is the **orbital angle** assigned to drone i , and θ_{base} rotates continuously at velocity ω_{orbit} to distribute suppression coverage. The **orbital radius** R_{orbit} adapts to fire size:

- $R_{\text{orbit}} = \alpha \cdot R_{\text{fire}}$

where α is a **scaling factor** (typically 0.7-1.0).

The swarm collectively approaches the target area, maintaining safe inter-drone spacing and dynamically adjusting to wind direction, fire spread rate, and terrain conditions. Communication between drones enables synchronized spray patterns, adaptive targeting, and redundancy handling if individual drones fail during the mission.

D. System Workflow

The proposed algorithm operates through six primary stages:

1. **Patrol Mode:** Scout drones perform continuous area monitoring
2. **Heat Detection:** A temperature anomaly triggers investigation mode
3. **Investigation Phase:** The drone moves to the heat source and collects thermal and visual data
4. **Fire Verification:** If the source is validated as a fire, a report is broadcast to the network
5. **Payload Deployment:** Swarm of firefighting drones is launched to the target site
6. **Suppression Phase:** Drones cooperatively disperse foam to extinguish the fire

E. Pseudocode of Swarm Logic

Initialize Drone_Swarm

For each drone in Scout_Drones:

 Begin Patrol_Mode ()

 If detect (heat_level > threshold):

 Investigate () confidence = ComputeFireConfidence ()

 If confidence > verification_threshold:

 Send_Coordinates ()

 Deploy_Payload_Drones ()

 break

For each drone in Payload_Drones:

 Navigate_To (Target_Coordinates)

 slot_angle = ComputeOrbitSlot (drone_id, total_drones)

Form_Swarm_Formation (slot_angle)

 Begin_Fire_Suppression ()

F. Decentralized Coordination

The system operates without a central controller through event-driven communication inspired by ACO's stigmergic coordination model. In ACO, ants deposit pheromones that guide subsequent ants without direct communication. Similarly, our system uses broadcast events that function as environmental markers triggering coordinated behaviours:

1. **FireSpotted**: Broadcast by scouts upon fire verification (analogous to pheromone deposition at food source)
2. **RespondToFire**: Global event triggering swarm deployment (analogous to recruitment signal)
3. **FireExtinguished**: Terminates suppression operations and releases drones (analogous to pheromone evaporation)

Each drone maintains its own state machine and responds to broadcasted events based on local decision rules. This architecture ensures continued operation even if individual drones or communication links fail—a critical requirement for GPS-denied environments where positioning systems may be unavailable, and communication may be degraded by smoke, structural interference, or damaged infrastructure.

G. Advantages

The system's hierarchical design allows for efficient division of labour—scouts handle detection while payload units focus on suppression. This approach minimizes false alarms, reduces latency, and enhances scalability. Additionally, the decentralized control model improves fault tolerance and enables real-time adaptability to environmental uncertainty.

The integration of PSO and ACO principles enables self-organization and robust operation in GPS-denied environments. The swarm maintains coordination through relative positioning and local communication without requiring external positioning systems. When individual drones fail, remaining agents reorganize autonomously through ACO-inspired stigmergic coordination.

IV. APPROACH

A. Implementation Details

The system is implemented in Unity 3D with C# scripting, providing a physics-based simulation environment. The architecture comprises the following key modules:

1. Drone Controller

Fig. 1 shows the Drone Controller class implementing the finite state machine for individual drone behavior with five states: IDLE, PATROL, RESPONDING, DELIVERING and RETURNING.

```
void Update()
{
    // Continuously adapt patrol altitude while patrolling
    if (state == DroneState.Patrol)
        _steer.desiredAltitude = ComputePatrolAltitude();

    switch (state)
    {
        case DroneState.Idle:
            _target = new Vector3(home.x, _steer.desiredAltitude, home.z);
            break;

        case DroneState.Patrol:
            if (!_target.HasValue ||
                (transform.position - _target.Value).sqrMagnitude < 100f)
            {
                PickNewPatrolPoint();
            }
            break;

        case DroneState.Responding:
            // Target set when responding. We just keep heading there / hovering.
            break;

        case DroneState.Delivering:
            UpdateOrbit();
            break;

        case DroneState.Returning:
            _target = new Vector3(home.x, _steer.desiredAltitude, home.z);
            if ((transform.position - _target.Value).sqrMagnitude < 36f)
            {
                state = DroneState.Idle;
            }
            break;
    }
}
```

Fig. 1. Drone States

The patrol altitude h_{patrol} adapts based on local building topology:

$$\bullet \quad h_{\text{patrol}} = \max(h_{\text{min}}, h_{\text{building}} + h_{\text{offset}})$$

where h_{building} represents the maximum building height within the drone's scan radius.

2. Steering Agent

Fig. 2 shows the SteeringAgent module implementing Reynolds flocking behaviors [1] with extensions for altitude control and obstacle avoidance:

```
public void TickSteering(Vector3 target, DroneController self)
{
    Vector3 force = Vector3.zero;

    // Altitude hold
    float altitudeError = desiredAltitude - transform.position.y;
    force += Vector3.up * altitudeError * altitudeForce;

    // Flocking
    var neighbors = Physics.OverlapSphere(transform.position, neighborRadius);
    Vector3 sep = Vector3.zero, all = Vector3.zero, coh = Vector3.zero;
    int count = 0;
    foreach (var col in neighbors)
    {
        if (!col.CompareTag("Drone")) continue;
        if (!col.attachedRigidbody) continue;
        count++;
        Vector3 toMe = transform.position - col.transform.position;
        float d = toMe.magnitude;
        if (d < 0.001f) continue;
        if (d < separationRadius) sep += toMe.normalized / Mathf.Max(d, 0.2f);
        all += col.attachedRigidbody.linearVelocity;
        coh += col.transform.position;
    }
    if (count > 0)
    {
        all = all.normalized;
        coh = ((coh / count) - transform.position).normalized;
    }
    force += separationWeight * sep + alignmentWeight * all + cohesionWeight * coh;

    // Wander
    force += Random.insideUnitSphere * wanderStrength;

    // Goal seeking
    if (target.HasValue)
    {
        Vector3 desired = (target.Value - transform.position);
        float dist = desired.magnitude;
        desired = desired.normalized * Mathf.Lerp(maxSpeed * 0.5f, maxSpeed, Mathf.Clamp01(dist / 10f));
        Vector3 seekForce = desired - rb.linearVelocity;
        force += Vector3.ClampMagnitude(seekForce, maxForce);
    }

    // Obstacle avoidance
    Vector3 fwd = transform.forward;
    Vector3 pos = transform.position;

    if (Physics.SphereCast(pos, avoidSphereRadius, fwd, out RaycastHit hit, avoidRange, obstacleMask, QueryTriggerInteraction.Ignore))
    {
        Vector3 away = Vector3.ProjectOnPlane(pos - hit.point, Vector3.up).normalized;
        if (away.sqrMagnitude < 1e-3f) away = Vector3.Cross(Vector3.up, fwd);
        force += away * avoidForce;
    }
    else
    {
        Vector3 leftDir = Quaternion.Euler(0f, -sideHiskerAngle, 0f) * fwd;
        Vector3 rightDir = Quaternion.Euler(0f, sideHiskerAngle, 0f) * fwd;
        if (Physics.Raycast(pos, leftDir, out hit, avoidRange * sideHiskerFactor, obstacleMask))
        {
            force += (Vector3.Cross(Vector3.up, leftDir)).normalized * avoidForce * 0.7f;
        }
        if (Physics.Raycast(pos, rightDir, out hit, avoidRange * sideHiskerFactor, obstacleMask))
        {
            force += (Vector3.Cross(rightDir, Vector3.up)).normalized * avoidForce * 0.7f;
        }
    }
}
```

Fig. 2. SteeringAgent altitude control and obstacle avoidance

3. FireSource

Fig. 3 shows the FireSource module simulating fire dynamics with intensity growth and suppression:

```
void Update()
{
    // how many payload drones are close enough on the XZ plane
    CurrentDronesInRange = CountPayloadDronesPlanar();

    if (CurrentDronesInRange >= dronesNeededToSuppress)
    {
        _unattendedTimer = 0f;
        _intensity -= suppressRatePerDrone * CurrentDronesInRange * Time.deltaTime;
    }
    else
    {
        _unattendedTimer += (CurrentDronesInRange == 0 ? Time.deltaTime : 0f);
        float rate = (_unattendedTimer >= unattendedRampSeconds) ? fastGrowRate : growRate;
        _intensity += rate * growthSpeed * Time.deltaTime;
    }

    // external spray suppression
    if (_externalSuppressionAccum > 0f)
    {
        _intensity -= externalSuppressionMultiplier * _externalSuppressionAccum * Time.deltaTime;
        _externalSuppressionAccum = 0f;
    }

    _intensity = Mathf.Clamp(_intensity, 0f, maxIntensity);

    // Concentric circles update
    if (useConcentricRings)
    {
        RebuildRings(); // ensure ring count matches intensity
        UpdateRingsVisual(); // size, width, color
    }

    // Extinguish if weak
    if (_intensity <= extinguishThreshold)
    {
        EventHub.FireExtinguished(this);
        Destroy(gameObject);
    }
}
```

Fig. 3. FireSource fire dynamics

4. SwarmManager

Fig. 4 shows the SwarmManager coordinates fleet deployment and event handling:

```
// ----- FIRE SPOTTED -----
void OnFireSpotted(FireSource fire, DroneController who, float estimate)
{
    if (fire == null) return;

    // 1. PAYLOAD AUTOLAUNCH (closest payload drones take off)
    if (payloadAutoLaunch)
    {
        var candidates = payloads
            .Where(p => p != null &&
                p.state != DroneController.DroneState.Delivering &&
                p.state != DroneController.DroneState.Responding)
            .OrderBy(p => (p.transform.position - fire.transform.position).sqrMagnitude)
            .ToList();

        if (!payloadAssignments.ContainsKey(fire))
        {
            _payloadAssignments[fire] = new List<DroneController>();
            var assignedList = _payloadAssignments[fire];

            int toTake = Mathf.Min(payloadsPerFire - assignedList.Count, candidates.Count);
            for (int i = 0; i < toTake; i++)
            {
                var p = candidates[i];
                assignedList.Add(p);

                // Assign orbit target. We KEEP slotting so they spread a little.
                int slotIndex = assignedList.Count - 1;
                p.SetOrbitTarget(fire, slotIndex, payloadsPerFire);
                // NOTE: we do NOT force their currentFire or state here, because
                // SetOrbitTarget already changes state to Delivering.
            }
        }
    }

    // 2. SEND A SCOUT RING (visual perimeter around fire/building)
    var scoutTeam = scouts
        .Where(s => s != null && s.state == DroneController.DroneState.Patrol)
        .OrderBy(s => (s.transform.position - fire.transform.position).sqrMagnitude)
        .Take(scoutsToDispatchPerFire)
        .ToList();

    float baseRadius = 12f;
    for (int i = 0; i < scoutTeam.Count; i++)
    {
        var s = scoutTeam[i];
        float angle = (360f / Mathf.Max(1, scoutTeam.Count)) * i + Random.Range(-12f, 12f);
        Vector3 dir = Quaternion.Euler(0f, angle, 0f) * Vector3.forward;
        Vector3 target = fire.transform.position + dir * baseRadius;

        s.AssignTarget(target);
        s.state = DroneController.DroneState.Responding;

        // force scouts to hover above the roofline
        float alt = Mathf.Clamp(
            fire.transform.position.y + s.attackOffset,
            s.minPatrolAlt,
            s.maxPatrolAlt
        );
        var steer = s.GetComponent<SteeringAgent>();
        if (steer != null)
        {
            steer.desiredAltitude = alt;
        }
    }
}

// ----- FIRE EXTINGUISHED -----
void OnFireExtinguished(FireSource fire)
{
    if (_payloadAssignments.ContainsKey(fire))
    {
        _payloadAssignments.Remove(fire);

        // Gently tell anyone mid-response to chill (optional)
        foreach (var p in payloads)
        {
            if (p != null && p.currentFire == fire)
            {
                p.ReturnAndIdle();
            }
        }

        foreach (var s in scouts)
        {
            if (s != null && s.state == DroneController.DroneState.Responding)
            {
                s.ReturnToPatrol();
            }
        }
    }
}
```

Fig. 4. SwarmManager coordinates fleet deployment

B. Sensor Simulation

Fig. 5 shows the DroneSensor module simulating thermal detection using sphere collision detection with dynamic beacon radius:

```

void OnTriggerStay(Collider other)
{
    if (_cooldown > 0f) return;
    if (!other.CompareTag("Fire")) return;

    var fire = other.GetComponent<FireSource>();
    var drone = GetComponent<DroneController>();
    if (!fire || !drone) return;

    float beacon = fire.BeaconRadius;
    float planar = Vector2.Distance(
        new Vector2(transform.position.x, transform.position.z),
        new Vector2(fire.transform.position.x, fire.transform.position.z)
    );
    if (planar > beacon) return;

    float proximity = 1f - Mathf.Clamp01(planar / beacon);
    float estimate = Mathf.Clamp01(fire.Intensity * (0.6f + 0.4f * proximity));

    EventHub.FireSpotted(fire, drone, estimate);
    _cooldown = reportCooldown;
}

```

Fig. 5. DroneSensor thermal detection

C. Foam Suppression Simulation

Fig. 6 shows the PayloadSprayer module implementing line-of-sight suppression with aiming constraints:

```

void LateUpdate()
{
    bool spray = false;
    Vector3 origin = nozzle ? nozzle.position : transform.position;

    if (ctrl.isPayloadDrone && ctrl.state == DroneController.DroneState.Delivering && ctrl.currentFire != null)
    {
        var fire = ctrl.currentFire;
        Vector3 tgt = fire.transform.position;
        tgt.y = Mathf.Clamp(tgt.y, ctrl.attackOffset, ctrl.minPatrolAlt, ctrl.maxPatrolAlt);

        Vector3 dir = tgt - origin;
        float dist = dir.magnitude;

        if (dist <= sprayRange && Vector3.Angle(transform.forward, dir) <= aimAngle)
        {
            if (!Physics.Raycast(origin, dir.normalized, dist, obstacleMask))
            {
                fire.AddExternalSuppression(suppressionPerSecond);
                lr.SetPosition(0, origin);
                lr.SetPosition(1, tgt);
                spray = true;
            }
        }
    }
    lr.enabled = spray;
}

```

Fig. 6. PayloadSprayer line-of-sight suppression

V. RESULT, ANALYSIS AND DISCUSSION

A. Simulation Environment

Experiments were conducted in a simulated urban environment measuring $500\text{m} \times 500\text{m}$ containing 30 buildings with heights ranging from 10m to 50m. The swarm comprised 10 scout drones and 10 payload drones distributed across 2 base stations.

Simulation Parameters:

- Scout patrol altitude: 12-150m (adaptive)
- Scout thermal sensor range: 18m base + 20m per intensity unit
- Payload spray range: 20m
- Payload suppression rate: 0.6 intensity units/second
- Fire growth rate: 0.06 units/s (unattended), 0.18 units/s (after 5s ramp)
- Maximum concurrent fires: 3

B. Detection Performance

The system demonstrated robust fire detection capabilities with the following metrics:

TABLE I. FIRE DETECTION PERFORMANCE

Metric	Value
Mean detection time	$3.2\text{s} \pm 1.1\text{s}$
False positive rate	0%
Detection success rate	100%
Scout response time	$1.8\text{s} \pm 0.6\text{s}$
Average confidence score	0.87 ± 0.09

The adaptive altitude patrol strategy enabled scouts to maintain effective thermal sensor coverage across varying building heights. Detection times varied based on scout proximity to fire spawn locations, with the distributed patrol pattern ensuring no area remained unmonitored for more than 5 seconds.

C. Suppression Effectiveness

Payload drone deployment and suppression coordination achieved the following results:

TABLE II. SUPPRESSION PERFORMANCE

Metric	Value
Mean detection time	$4.7\text{s} \pm 1.3\text{s}$
Payload drones per fire	4.2 ± 0.8
Mean suppression time	$12.4\text{s} \pm 3.6\text{s}$
Fire extinguish success rate	98%
Collision avoidance success	100%

The PSO-inspired swarm coordination algorithm successfully maintained collision-free orbital formations while delivering suppression foam. The adaptive orbital radius adjustment prevented drones from clustering excessively while ensuring all drones remained within effective spray range.

D. Scalability Analysis

System performance was evaluated with varying fleet sizes:

TABLE III. SCALABILITY METRICS

Scouts	Payloads	Coverage Area	Mean Response Time
5	5	180,000 m ²	5.9s
10	10	250,000 m ²	3.2s
20	20	400,000 m ²	2.1s

Results demonstrate near-linear scaling in coverage area with fleet size, while response times improve due to higher probability of scout proximity to fire events. The decentralized architecture exhibited no performance degradation with increased fleet size, confirming scalability.

E. Robustness to Failures

System resilience was tested by randomly disabling drones during operations:

TABLE IV. FAILURE RESILIENCE

Failure Scenario	Mission Success Rate
1 scout disabled	100%
2 payload disabled	95%
3 drones disabled (mixed)	92%
Base station offline	92%

The decentralized control architecture enabled continued operation despite individual drone failures. When payload drones fail during suppression, remaining drones automatically redistributed orbital positions through PSO-inspired self-organization to maintain coverage. Scout failures reduced detection coverage but did not prevent fire responses by remaining scouts, demonstrating the ACO-inspired stigmergic coordination's robustness.

F. Comparison with Baseline Approaches

The swarm system was compared against alternative architectures:

TABLE V. FAILURE RESILIENCE

Approach	Detection Time	Suppression Time	Success Rate
Single drone patrol	12.3s $\pm$ 4.2s	N/A (no suppression)	N/A
Centralized multi-drone	4.1s $\pm$ 1.5s	15.8s $\pm$ 4.9s	85%
Proposed swarm	3.2s $\pm$ 1.1s	12.4s $\pm$ 3.6s	98%

The proposed swarm approach achieved faster detection through distributed patrol coverage and faster suppression through parallel deployment of multiple payload drones. The centralized approach exhibited higher failure rates due to communication bottlenecks and single-point vulnerabilities—issues that the bio-inspired decentralized architecture avoids.

VI. POSSIBLE USAGE AND APPLICATIONS

The proposed autonomous drone swarm system demonstrates versatility across multiple real-world emergency response and monitoring scenarios. This section outlines potential deployment contexts and application domains.

A. Urban Fire Response

The main purpose of the application is to respond quickly to building fires in crowded metropolitan settings. Scout drones can be used to continuously monitor high-risk areas from fire stations or other municipal facilities. When a fire is detected, payload drones provide immediate suppression to stop the fire from spreading until human firefighters are deployed. This could stop room fires from becoming structural fires.

Key advantages for urban deployment include:

- 24/7 automated monitoring without human operator fatigue
- Rapid response times (under 5 seconds from detection to deployment)
- Access to rooftops and upper floors difficult for ground crews
- Reduced risk to human responders during initial assessment and suppression

B. Wildfire Early Detection and Containment

For wildland-urban interface zones and forested areas, scout drones can patrol large territories to detect fires during their initial stage. Early detection enables payload drones to suppress small fires before they develop into major wildfires, potentially preventing catastrophic losses.

The GPS-denied operation capability is particularly valuable in remote wilderness areas where satellite positioning may be unreliable due to terrain or atmospheric conditions. The swarm can maintain coordination through relative positioning and local communication.

C. Industrial Facility Monitoring

Chemical plants, oil refineries, and manufacturing facilities face elevated fire risks due to flammable materials and complex equipment. Permanent drone installations can provide continuous thermal monitoring of critical infrastructure, with payload drones equipped with specialized suppression agents appropriate for chemical fires.

The decentralized architecture ensures continued operation even if facility communication systems are damaged during an incident. Drones can operate autonomously within the facility perimeter using visual landmarks for navigation.

D. Port and Maritime Infrastructure

Shipping ports, container yards, and maritime facilities present unique firefighting challenges due to limited access routes and high-value cargo. Scout drones can monitor extensive port facilities, while payload drones can reach fires on vessels or in storage areas inaccessible to ground equipment.

The system's ability to operate in communication-uncertain environments is valuable in port settings where large metal structures may interfere with radio signals.

E. High-Rise Building Surveillance

Tall buildings pose significant challenges for conventional firefighting due to limited exterior access and long vertical travel distances. Permanently installed drone systems can patrol high-rise exteriors and respond to fires on upper floors before internal spread compromises evacuation routes.

The adaptive altitude control enables drones to operate effectively across the full height range of modern skyscrapers while avoiding obstacles such as architectural features and window-washing equipment.

F. Airport and Aircraft Facilities

Aircraft hangars and airport facilities contain high-value assets and aviation fuel, creating severe fire hazards. The swarm system can provide rapid response to contain fires before they spread to adjacent aircraft or fuel storage, potentially preventing multi-million-dollar losses.

The collision avoidance and formation control capabilities enable safe operation in the complex obstacle environment of airport facilities with multiple buildings, aircraft, and ground vehicles.

VII. CONCLUSION AND FUTURE WORK

This paper presented an autonomous drone swarm framework for fire detection and suppression. By integrating decentralized swarm intelligence algorithms adapted from Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO) principles, the system enables drones to self-organize, distribute tasks dynamically, and maintain robust communication in uncertain and GPS-denied environments typical of disaster scenarios.

The system integrates scout drones for early detection and payload drones for rapid firefighting deployment. The proposed decentralized approach enables autonomous coordination, efficient task distribution, and resilience against individual drone failure or communication degradation. Experimental results demonstrate the system's effectiveness with mean detection times under 3.5 seconds and successful suppression of 98% of fires within 12.4 seconds of detection.

Key innovations include adaptive altitude patrol strategies for scouts, PSO-inspired obstacle-aware orbital formation control for payload drones, and a fully decentralized ACO-inspired coordination architecture through stigmergic event communication. The system maintains coordination through relative positioning and local communication without requiring GPS or centralized control, ensuring continued operation when positioning systems are unavailable and communication is degraded by smoke, structural interference, or damaged infrastructure.

Future work includes:

1. **GPS-Denied Navigation:** Integration of visual-inertial odometry (VIO) and simultaneous localization and mapping (SLAM) for operation in environments where GPS is unavailable or unreliable
2. **Enhanced Communication Protocols:** Development of mesh networking with multi-hop relay capabilities to extend communication range in smoke-filled or structurally complex environments
3. **Adaptive Swarm Algorithms:** Implementation of more sophisticated bio-inspired algorithms that adjust PSO/ACO parameters in real-time based on environmental conditions
4. **Hardware Implementation:** Real-world flight trials with physical drone platforms
5. **Predictive Modelling:** Optimizing foam dispersion strategies under dynamic environmental conditions using machine learning

The system demonstrates significant potential as a scalable, cost-effective, and reliable asset for next-generation emergency response operations capable of operating effectively in the most challenging and uncertain environments.

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